

Topic: Converting ER Model to Tables (Relational Database)

Q. How is an ER Diagram mapped into a Relational Database schema? Explain the rules for converting Entities, Attributes, and the various Relationships (1:1, 1:N, M:N) into database tables.

Definition to Remember

ER-to-Relational Mapping is the process of converting a conceptual ER diagram into a physical Relational Model consisting of Tables, Columns, and Keys. Each entity, attribute, and relationship must be transformed using a specific set of mapping rules.

1. Rule 1: Strong Entities and Attributes

- Every **Strong Entity Type** → becomes a separate **Table**.
- Simple Attributes of the entity → become **Columns**.
- The **Key Attribute** → becomes the **Primary Key** of the table.
- **Composite Attributes**: Only the sub-components become columns (e.g., **Address** → **Street** , **City** , **Zip**).

2. Rule 2: Multi-Valued Attributes

- Cannot store a multi-valued attribute in the main table (relational tables require **atomic** values).
- **Solution**: Create a **new separate table** with:
 - The entity's Primary Key (as a **Foreign Key**)
 - The multi-valued attribute
 - Both together form the **Primary Key** of this new table.

3. Rule 3: Weak Entities

- Weak Entity → converted into a **separate table** with its own attributes.
- Add the **Primary Key of the Identifying Strong Entity** as a **Foreign Key**.
- The new table's PK = **FK + Partial Key** (discriminator) of the weak entity.

4. Rule 4: Mapping Binary Relationships

Cardinality	New Table Needed?	Action
1:1	No	Add PK of one table as FK in the other (prefer total participation side)
1:N	No	Add PK of the "One" side as FK in the "Many" side table
M:N	Yes	Create a Junction Table with both PKs as FKs; their combination = new PK

1:1 Mapping:

MANAGER (Mgr_ID) — DEPARTMENT (Dept_ID, Mgr_ID as FK)

1:N Mapping:

DEPARTMENT (Dept_ID) — EMPLOYEE (Emp_ID, Dept_ID as FK)

M:N Mapping:

STUDENT (Roll_No) — ENROLLS (Roll_No FK, Course_ID FK) — COURSE (Course_ID)

★ Must-Write Points (for 10 marks)

1. ER-to-Relational Mapping converts ER blueprints into physical Tables, Columns, and Keys.
2. Every Strong Entity → one Table; its Key Attribute → Primary Key.
3. Multi-Valued Attributes cannot exist in the main table — a new table is created.
4. Weak Entity → new table; PK = Owner Entity's PK (FK) + Partial Key.
5. 1:1 relationship → no new table; place one PK as FK in the other.
6. 1:N relationship → no new table; place "One" side's PK as FK in "Many" side.
7. M:N relationship → **requires a Junction (Associative) Table** with both PKs as FKs.

⚡ Quick Recall

Strong Entity → Table (PK) → Multi-valued → New Table → Weak Entity → Table (FK + Partial Key) →
1:1 (FK) → 1:N (FK on Many) → M:N (Junction Table)